

FROM 143.7 MILLION PARAMETERS TO 1,893 WEIGHTS

A physics-informed path toward compact, event-driven, MCU-ready motor diagnosis

This report places three reductions on the same quantitative trajectory: large deep networks from the literature, the compact recurrent model developed in the MCSA study, and the current hard-forward spiking neural network. Model size, compute load, memory, and estimated energy are treated as separate quantities.



Central result: the main advance is not compression alone. A physically aligned signal representation first removes unnecessary model capacity. Event-driven inference then removes most of the remaining dense weighted computation.

Reference figures

Stage	Architecture	Parameters	Accuracy	Primary result
Literature	VGG-19	143.7 M	99.4%	High accuracy, GPU-scale cost
MCSA paper	LSTM h=32	4,773	88.0%	19.1 KB float32
SpikyPanda experiment	GRU h=32	3,621	91.0% test	364/400
SpikyPanda experiment	SNN 32+5 LIF	1,893	78.5% test	314/400, grouped split

Terminology note: the million-parameter models in the MCSA paper are deep CNN architectures, mainly VGG-19 and NASNet-Mobile, rather than conventional MLPs. The explicitly evaluated MLP is a 773-parameter FFT+MLP baseline that reaches 67.0%.

1. Why this comparison matters

Accuracy alone does not establish deployability. A 143.7-million-parameter network can reach 99.4%, yet its float32 weights require approximately 574.8 MB and the cited execution environment is GPU-based. An MCU-oriented solution must fit in kilobytes while limiting memory traffic, multiplications, nonlinear functions, and active time.

The research question is therefore stricter than peak accuracy: **what is the smallest representation and compute budget that can retain useful broken-rotor-bar severity information?**

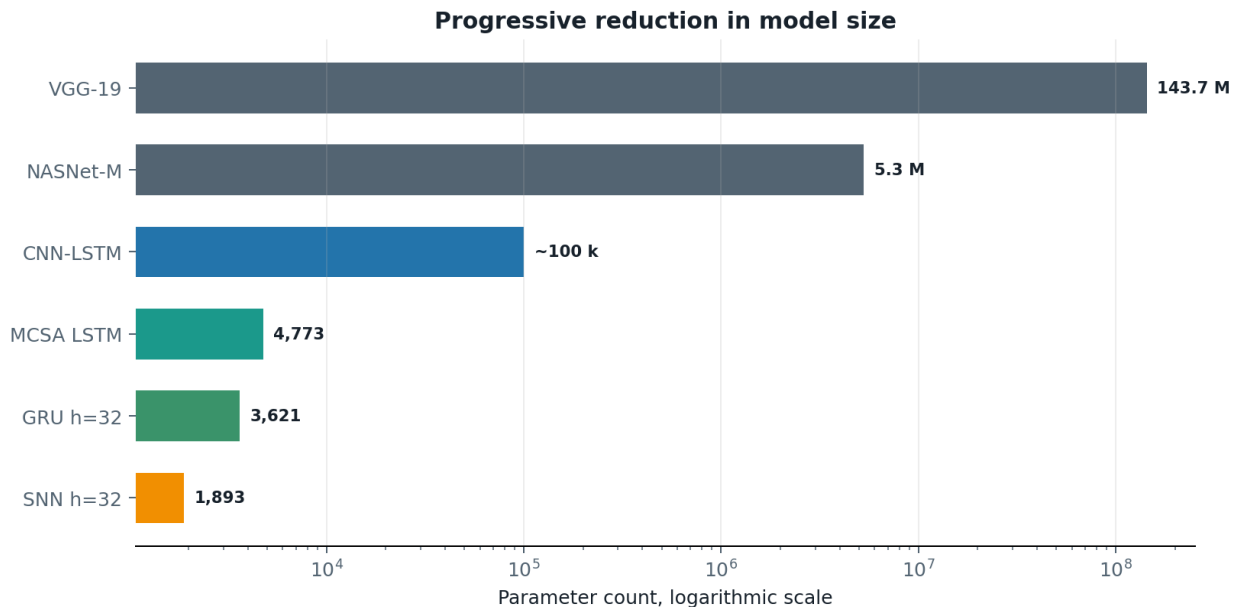


Figure 1. Parameter counts for the cited architectures and the SpikyPanda models. A logarithmic axis is required to make the compact models visible.

Correct interpretation of the reduction factors

Comparison	Ratio	Meaning
VGG-19 / SNN	$143,700,000 / 1,893 = 75,909x$	Parameter count reduction
NASNet-M / SNN	$5,300,000 / 1,893 = 2,800x$	Parameter count reduction
MCSA LSTM / SNN	$4,773 / 1,893 = 2.52x$	Parameter count reduction
GRU / SNN	$3,621 / 1,893 = 1.91x$	Parameter count reduction
GRU compute / SNN propagation	$225,280 / 5,305 = 42.47x$	Weighted operation reduction

The factor close to 45 does not describe GRU-to-SNN model size. It describes weighted propagations per observed window. Weight storage is reduced by 1.9x.

2. First breakthrough: understand the signal before scaling the network

The MCSA paper starts from the fault physics. A broken rotor bar creates sidebands around the supply frequency. For supply frequency f and slip s , the principal components occur near $f(1 - 2s)$ and $f(1 + 2s)$. In the time domain, they appear as a slow amplitude modulation, typically between 2 and 6 Hz on the studied test bench.

Approximate modulation frequency: $f_{\text{mod}} = 2 s f$

Large CNNs operating on raw current or spectrograms have enough capacity to learn an implicit envelope detector. The MCSA work makes that operation explicit with a half-cycle moving RMS:

$$\text{env}(t) = \sqrt{\frac{1}{W} \sum_{i=0}^{W-1} x(t+i)^2}$$

After transient removal, decimation to approximately 60 Hz, and per-window centering, 256 high-frequency values are replaced by 64 envelope samples. The slow fault signature becomes directly accessible to the recurrent network.

Ablation reported in the paper

Representation	Accuracy	Observed failure or gain
Raw, per-trace normalization, 64 steps	26.8%	Amplitude normalization removes the signature
Raw, global normalization, 256 steps	16.0%	Insufficient recurrent gradient through the long sequence
Envelope, global min/max	35.0%	Load level remains dominant
Centered envelope	88.0%	The useful modulation becomes visible

The decisive gain is not another layer. It is the removal of irrelevant carrier structure and the load-dependent baseline. This physics-informed representation reduces the capacity requirement by several orders of magnitude.

What the small MLP demonstrates

The FFT+MLP baseline contains only 773 parameters, fewer than the current SNN, but reaches 67.0%. A small parameter count is therefore not sufficient. The architecture must match the temporal structure of the phenomenon. The LSTM, GRU, and SNN each provide a form of memory that the small MLP lacks.

MCSA paper result

The LSTM $h=32$ contains 4,773 parameters, requires 19.1 KB in float32, and reaches 88.0% on five severity levels. Binary healthy/fault accuracy reaches 97.3%. The result establishes that explicit physical representation can replace a large fraction of deep-network capacity.

3. Second breakthrough: from LSTM to a compact GRU

The GRU retains recurrent memory while using three weight groups instead of the four LSTM gates. With 3 inputs, 32 hidden units, and 5 outputs, the exact count is:

GRU input: $3 \times 3 \times 32 = 288$ GRU recurrence: $3 \times 32 \times 32 = 3,072$ GRU biases: $3 \times 32 = 96$ Output: $32 \times 5 + 5 = 165$ Total: 3,621 parameters

The best recorded checkpoint reaches 93.5% validation accuracy and 91.0% on 400 test windows, with 364 correct classifications. Its weights occupy exactly 14,484 bytes in float32, or 14.14 KiB.

Relative to the paper's LSTM, the GRU reduces parameter count by 24.1% while moving from 88.0% to 91.0% on the observed test. This is encouraging but not a controlled ablation, because the protocol and pipeline version also changed.

GRU compute cost

The current runtime computes all three GRU weight groups and five outputs at each of 64 time steps:

MACs per step = $3 \times 32 \times (3 + 32) + 32 \times 5 = 3,520$ MACs per window = $3,520 \times 64 = 225,280$

The runtime also evaluates 6,464 sigmoid or tanh functions. On an MCU, these can cost more than a multiply-accumulate unless replaced by lookup tables or dedicated approximations.

Accuracy and compactness

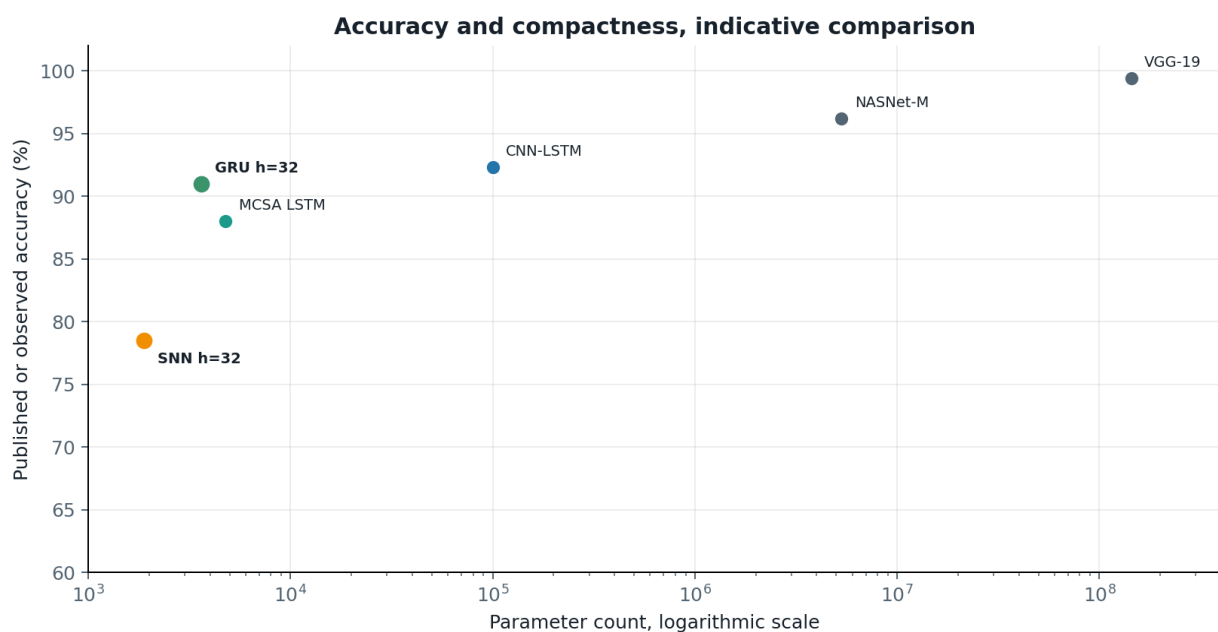


Figure 2. Each point must be interpreted with its experimental protocol. Literature results, the historical GRU, and the grouped SNN do not all use the same acquisition split.

4. Third breakthrough: from dense computation to event-driven computation

The current SNN does not simply replace GRU activations with zeros and ones. It changes the interface between the signal and the network. Nine sensor cells filter the three current envelopes around three frequency fields. Phase crossings and three intensity levels produce 54 binary spike ports.

Compiled topology: observation -> 9 IIR cells -> 54 ports -> 32 hidden LIF -> 5 class LIF

Exact weight count

Sensor to hidden: $54 \times 32 = 1,728$ Hidden to classes: $32 \times 5 = 160$ Frame-end signal to classes: 5 Total: 1,893 trainable weights

The training forward pass is strictly identical to native LIF inference: binary thresholding, hard reset, and no fractional values transmitted over neural links. The surrogate derivative is used only for gradient computation. The loss is aligned with the score read by the deployed runtime:

$\text{score}[c] = 2 \times \text{spike_count}[c] + \text{final_membrane}[c] / \text{threshold}[c]$

This objective change raised the best hard-forward SNN from 51.0% to 78.5% on the independent test while reducing activity from 179.2 to 118.6 neuron spikes per window.

Measured event-driven work

Source	Computation	Average weighted operations
Sensor -> 32 LIF	147.24 events x 32	4,712 weight additions
32 LIF -> 5 classes	117.64 hidden spikes x 5	588 weight additions
Frame end	1 event x 5	5 weight additions
Network total		approximately 5,305 additions

Because spike amplitude is exactly 1, synaptic propagation can reduce to **membrane += weight**. Multiplications remain in the nine IIR filters and membrane decay, but the full dense matrix is no longer evaluated at every time step.

The measured window requires 225,280 GRU MACs versus approximately 5,305 SNN synaptic additions, a 42.5x reduction in weighted propagation.

5. Memory, CPU load, and estimated energy

Weight storage

Model	Parameters	Float32 weights	Int8 weights	Context
VGG-19	143.7 M	574.8 MB	143.7 MB	GPU in cited work
NASNet-M	5.3 M	21.2 MB	5.3 MB	GPU in cited work
CNN-LSTM	~100 k	~400 KB	~100 KB	Deep hybrid
MCSA LSTM	4,773	19.1 KB	4.8 KB	MCSA paper
GRU h=32	3,621	14.5 KB	3.6 KB	Observed checkpoint
SNN h=32	1,893	7.6 KB	1.9 KB	Weights only

SNN deployment must also store coefficients for nine filters, thresholds, LIF constants, and possibly routing indices. A specialized float32 layout is estimated at 8 to 9 KiB. A generic sparse graph is more likely to require 10 to 13 KiB. With int8 weights, routing metadata can cost more than the weights themselves, so dense blocks should compile into contiguous arrays.

Compute load

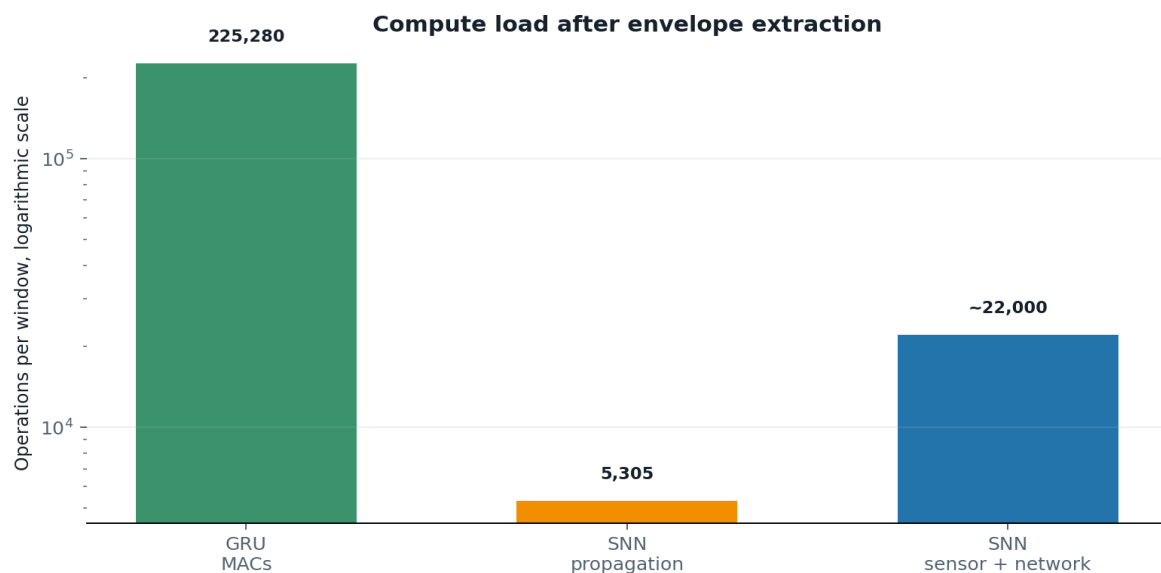


Figure 3. The SNN sensor adds 1,152 biquad updates. Including filters, thresholds, LIF dynamics, and routing, useful arithmetic is estimated at 17,000 to 30,000 simple operations per window.

Illustrative MCU estimate, model only

Illustrative assumption	GRU	Specialized SNN
Cycles per window	0.5 to 1.5 million	50,000 to 150,000
80 MHz MCU	6.25 to 18.75 ms	0.63 to 1.88 ms
At 10 mW active power	approximately 60 to 190 μ J	approximately 6 to 20 μ J
Expected energy ratio	reference 1	approximately 5x to 15x lower

The microjoule values are not measurements. They illustrate a plausible range. The ratio is more robust than the absolute value, which depends on the MCU, numeric format, memory, and compiler.

6. What the estimate includes, and what it excludes

Shared envelope cost

The current GRU and SNN consume a precomputed RMS envelope. On the target device, this envelope must be produced from three current channels sampled near 55.6 kHz. A 1.066-second window contains close to 60,000 samples per phase, approximately 178,000 squares, and 356,000 additions or subtractions for a basic moving RMS.

If the same MCU performs this stage, it becomes a significant shared cost and narrows the end-to-end energy advantage to an estimated 1.5x to 3x. If a sensor, DSP, or existing acquisition stage provides the envelope, the neural-engine advantage remains closer to 5x to 15x.

Software representation matters

The current SNN checkpoint takes approximately 13.5 ms per window in the browser. This does not represent MCU potential. The TypeScript runtime uses a dynamic graph, event objects, channels, and generic exponentials. The projected gain requires contiguous arrays, implicit fanouts, no allocation per spike, and precomputed decay coefficients.

Working RAM

Element	GRU h=32	SNN with 9 sensors and 37 LIF
Primary state	32 hidden values and gate buffers	9 IIR states and 37 membranes
Estimated optimized RAM	0.6 to 1.2 KiB	1 to 3 KiB
Primary risk	Buffers not reused	Event queues and graph metadata

The SNN clearly reduces weights and weight traffic, but it does not automatically reduce RAM. An overly generic event scheduler can use more state than a compact GRU kernel.

Accuracy comparability

The 91.0% GRU uses the historical dataset fingerprint dcb578a0 with 64 steps. The 78.5% SNN uses fingerprint ec00f5c3, 128 steps at 120.110 Hz, and an acquisition-grouped split. The grouped protocol prevents an acquisition, or overlapping windows from that acquisition, from appearing in both training and testing.

It would therefore be incorrect to claim an exact 12.5-point loss for the SNN under identical conditions. A scientific comparison requires training the GRU h=32 on ec00f5c3 with the same training, validation, and 400 independent test windows.

Load-stratified robustness

The test set covers eight load levels from 12.5% to 100%. The aggregate checkpoint confusion matrix does not identify the loads at which errors occur. The paper must therefore avoid attributing Healthy/BRB1 errors to light load until inference is replayed and every prediction is linked to its source acquisition.

Measurement to report at each load	Scientific purpose
Window count and accuracy	Verify representativeness and overall stability
Healthy recall and BRB1 recall	Locate the detection limit for mild faults
Healthy to BRB1 and BRB1 to Healthy errors	Separate false alarms from missed faults
Classification margin and event rate	Relate uncertainty to physical signal visibility

7. R&D interpretation

Observed progression

Stage	Question addressed	Result
Large deep networks	Can very high accuracy be reached?	Yes, 96% to 99.4%, with 5.3 M to 143.7 M parameters
MCSA preprocessing	What physical information is actually useful?	2 to 6 Hz envelope, 4,773-parameter LSTM
Compact GRU	Can recurrent memory be reduced?	3,621 parameters, 91.0% on historical test
Hard-forward SNN	Can dense computation become event-driven?	1,893 weights, 5,305 propagations, 78.5% grouped test

What the SNN already demonstrates

The current SNN demonstrates that a network with strictly binary neural communication can learn five-level classification without forward pseudo-spikes. It reaches 78.5% on an independent grouped test using 1,893 weights, 147.2 sensor events, and 118.6 neuron spikes per window. Its main advantage is not yet accuracy. It is the transition from systematic dense evaluation to signal-conditioned activity.

What remains to be demonstrated

Validation	Required measurement
GRU on grouped split	Accuracy, MACs, time, and energy on ec00f5c3
Compact MCU SNN kernel	Exact cycles, flash, RAM, and absence of allocation
Quantization	Float32 accuracy compared with int8/int16
Physical energy	μJ per window measured with a shunt or power analyzer
Complete preprocessing	Separate shared RMS, wave sensor, and network costs
Robustness by load	Accuracy, Healthy/BRB1 recall, margin, and event rate for each of the eight loads

Conclusion: this trajectory does not show that an SNN is merely a smaller network. Physics-informed representation first removes millions of unnecessary parameters. Event-driven representation then removes approximately 42x of the remaining weighted propagation. The architecture is credible for MCU deployment, provided that its cost is confirmed on a compact embedded kernel.

References and data provenance

[1] G. Pelletier, *Envelope-Domain Preprocessing for Ultra-Compact LSTM-Based Broken Rotor Bar Severity Grading*, local document motor-current-mcsa-paper.pdf, 5 pages.

[2] GRU checkpoint motor_current_best_val_93p5.json: GRU h=32, 3,621 parameters, 93.5% validation, 91.0% test (364/400), fingerprint dcb578a0.

[3] SNN checkpoint motor_current_snn_h32_5a83490c_best_val_82p5.json: 1,893 weights, 82.5% validation, 78.5% test (314/400), fingerprint ec00f5c3.

[4] Local experiment report runtime-decoder-loss-experiment.md: hard-forward protocol, neural activity, native runtime identity, and loss comparison.

[5] VGG-19, NASNet-Mobile, and CNN-LSTM figures are reproduced from the MCSA paper's comparison table, which cites Barrera-Llana et al. and Jakaria et al. Published protocols are not necessarily identical.